15/01/2021

Aakash
Medical | IIT-JEE | Foundations
***Regd. Office*: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph. 011-47623456**

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 5

Topics covered :

Physics **:** Thermodynamics, Kinetic Theory, Oscillations and Waves

Chemistry **:** Organic Chemistry: Some Basic Principles and Techniques, Hydrocarbons, Environmental Chemistry

Botany **:** Mineral Nutrition, Photosynthesis in Higher Plants, Respiration in Plants

Zoology **:** Locomotion & Movement, Neural Control & Coordination

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. One mole of an ideal monoatomic gas is isobarically expanded from volume V to $2V$. If initial temperature of gas is T, then the increase in internal energy of gas is

   (1) $2RT$

   (2) $\frac{3RT}{2}$

   (3) RT

   (4) $\frac{5}{2}RT$

2. Three moles of oxygen at 400 K and one mole of ammonia at 300 K are mixed in an isolated fixed chamber. The final temperature of the mixture is

   (1) 391.4 K

   (2) 210.6 K

   (3) 371.4 K

   (4) 310.5 K

3. 5 moles of an ideal gas are compressed to half the initial volume at constant temperature of 300 K. The work done by the gas in the process is (R = 8.3 J K^{-1} mol^{-1})

   (1) 4.2×10^3 J

   (2) -2.1×10^4 J

   (3) -8.6×10^3 J

   (4) -6.3×10^3 J

4. The kinetic molecular theory of gases predicts that at a given temperature
   (1) Molecules of all gases have the same average translational kinetic energy per molecule
   (2) Lighter gas molecules have lower average translational kinetic energy per molecule than heavier gas molecule
   (3) All molecules in a gas don't have same average translational kinetic energy per molecule
   (4) Lighter gas molecules have higher average translational kinetic energy per molecule than heavier gas molecules

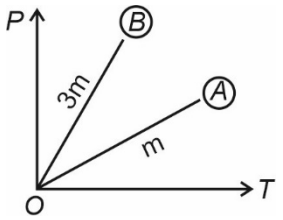
5. $\Delta U = 0$ in a non cyclic process of an ideal gas. The process
   (1) May be isobaric
   (2) May be isochoric
   (3) May be isothermal
   (4) Must be isothermal

6. A certain amount of heat energy is supplied to a monoatomic ideal gas which expands at constant pressure. What fraction of heat energy is converted into work?
   (1) $\frac{2}{5}$ (2) $\frac{2}{3}$
   (3) $\frac{5}{7}$ (4) $\frac{3}{5}$

7. In an adiabatic compression the product of pressure and volume
   (1) Decreases
   (2) Increases
   (3) Remains same
   (4) First increases then decreases

8. In a process, the pressure of an ideal gas is proportional to square of volume of the gas. If the temperature of the gas increases, in this process the work done by the gas is
   (1) Negative
   (2) Positive
   (3) Zero
   (4) May be positive or zero

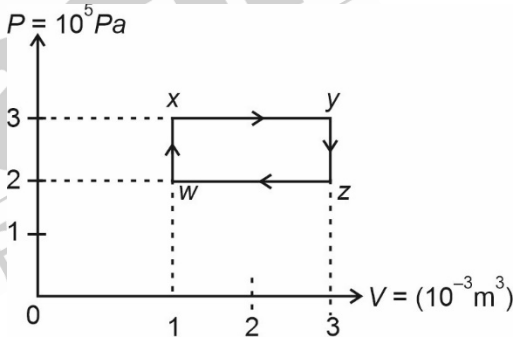
9. Two different masses m and $3m$ of a perfect gas are heated separately in a vessel of constant volume, the pressure and absolute temperature graphs for these cases are shown as A and B. The ratio of slopes of curve B to A is



   (1) $\frac{2}{3}$ (2) 3
   (3) $\frac{1}{3}$ (4) $\frac{1}{2}$

10. Two moles of an ideal gas monoatomic in nature occupies a volume V at 27°C. The gas is expanded adiabatically to a volume of $8V$. Final temperature of the gas is
   (1) 20°C (2) 5°C
   (3) 200 K (4) 75 K

11. A gas undergoes the cycle of pressure and volume changes as shown in figure. The net work done by the gas



   (1) 100 J
   (2) – 100 J
   (3) 200 J
   (4) – 400 J

12. Efficiency of Carnot's engine is 50% when temperature of sink is 500 K. In order to increase efficiency to 60% keeping temperature of source same, what should be temperature of sink?
   (1) 400 K
   (2) 500 K
   (3) 600 K
   (4) 650 K

13. 100% conversion of heat into work is impossible even when the engine used is ideal. It implies from

(1) First law of thermodynamics

(2) Second law of thermodynamics

(3) Zeroth's law of thermodynamics

(4) Kinetic theory of gases

14. For a certain apparatus the diffusion rate of hydrogen gas has an average value of 28.7 cm^3s^{-1}, the diffusion of another gas under the same condition is measured to have an average rate of 7.2 cm^3s^{-1}. The gas is

(1) Carbon dioxide (2) Helium

(3) Neon (4) Oxygen

15. Two gases are under identical condition of temperature, pressure and volume. The ratio of mean free paths of molecules of two gases having molecular diameter 2 Å and 3 Å respectively is

(1) 1.5 (2) 6

(3) 0.44 (4) 2.25

16. The equation of displacement of a particle in SHM is given by $y = \sin\omega t + \cos\omega t$. Then the ratio of amplitude to magnitude of initial displacement is

(1) 1 (2) $\sqrt{2}$

(3) $\sqrt{3}$ (4) 2

17. The time period of a SHM is 6 s and amplitude is 6 cm. The displacement from mean position at a time 8 s after the particle passing through its right extreme position is

(1) – 2 cm (2) 3 cm

(3) – 3 cm (4) – 4 cm

18. In SHM, projection of uniform circular motion of a uniformly revolving particle on a circle, having sense of rotation clockwise can be given by relation $x = 2\sin\left(2t + \frac{\pi}{3}\right)$ where x is in cm and t in second. The initial phase (in rad) and time period of revolution of particle will be

(1) $\frac{\pi}{3}, \pi$ s (2) $\frac{\pi}{3}, 2\pi$ s

(3) $-\frac{\pi}{3}, 2\pi$ s (4) $-\frac{\pi}{3}, \pi$ s

19. A platform performs simple harmonic oscillations in a vertical plane. The amplitude of oscillation is 20 cm. A coin placed on top of platform is not separated from it, the least time of these oscillation is (g = 9.8 ms^{-2})

(1) 0.3 s (2) 0.6 s

(3) 0.9 s (4) 1.2 s

20. Which of the following is not characteristics of simple harmonic motion?

(1) The motion is periodic

(2) In oscillations energy is conserved

(3) The motion is along a straight line about equilibrium position

(4) The acceleration of the particle is directed towards extreme position

21. In simple harmonic motion, the velocity of particle is given by $v = \sqrt{144 - 16x^2}$ m s^{-1}, where x is in meter. The maximum value of particle velocity during motion is

(1) 6 m s^{-1} (2) 16 m s^{-1}

(3) 12 m s^{-1} (4) 13 m s^{-1}

22. A particle is in simple harmonic motion. At the extreme position, the force acting is 10 N. The force on the particle at a point midway between mean and extreme position is

(1) 2 N (2) 2.5 N

(3) $5\sqrt{2}$ N (4) 5 N

23. Natural length of a spring is 30 cm and its spring constant is 3000 N m^{-1}. A mass of 10 kg is slowly hung from it. The time period of oscillation of this system is

(1) $\frac{\pi}{5}$ s (2) $\frac{\pi}{5\sqrt{3}}$ s

(3) $\frac{\pi}{10\sqrt{3}}$ s (4) $\frac{\pi}{2}$ s

24. A light spring supports a 500 gram weight at its lower end. It oscillates up and down with a period of 4 s. How much weight must be removed from lower end of spring to reduce period to 2 s?

(1) 275 gm

(2) 325 gm

(3) 375 gm

(4) 125 gm

25. For a simple harmonic oscillator, potential energy is equal to kinetic energy
    (1) Once during each cycle
    (2) Twice during half cycle
    (3) When $x = A$ (amplitude)
    (4) At each moment of time

26. A small mass body attached to one end of a spring oscillates horizontally with SHM, with frequency of $\frac{1}{2\pi}$ (Hz) and total energy of 40 J. If maximum speed is 0.8 m s^{-1}, force constant of the spring is
    (1) 250 N m^{-1}
    (2) 500 N m^{-1}
    (3) 125 N m^{-1}
    (4) 1000 N m^{-1}

27. For a simple pendulum, graph between time period and square root of its length will be
    (1) A straight line (2) A parabola
    (3) A circle (4) A hyperbola

28. For a damped oscillator the mass of the block is 200 g. Spring constant is 90 N m^{-1} and damping constant is 40 g s^{-1}. The period of oscillation is approximately
    (1) 0.3 s (2) 0.4 s
    (3) 0.5 s (4) 0.9 s

29. The potential of a particle of mass 10 g placed in a potential field is given by $V = (50x^2 + 10)$ erg/g. The frequency of oscillation is
    (1) $\frac{2}{\pi}$ s^{-1} (2) $\frac{3}{\pi}$ s^{-1}
    (3) $\frac{5}{\pi}$ s^{-1} (4) $\frac{6}{\pi}$ s^{-1}

30. Transverse waves are not possible inside fluids because
    (1) Fluids have low density
    (2) Fluids yield to shear forces
    (3) Fluids have infinite shear modulus
    (4) Both (2) and (3) are correct

31. A travelling waves passes a point of observation. At this point the time interval between successive crests is 0.2 s and separation between adjacent crests is 2 m, then
    (1) Wavelength of wave is 1 m
    (2) Velocity of propagation is 20 m s^{-1}
    (3) Wave frequency is 5 Hz
    (4) Wave frequency is 10 Hz

32. A simple harmonic progressive wave is represented by an equation $y = 8 \sin 2\pi(0.1x - 2t)$ where x and y are in cm and t in second. At any instant phase difference between two particles separated by 1 cm along x-direction is
    (1) 36° (2) 45°
    (3) 72° (4) 90°

33. In a standing wave, all the particles of medium
    (1) Have zero displacement simultaneously at some instant
    (2) Are at rest simultaneously at some instant
    (3) Have maximum displacement simultaneously at some instant
    (4) All of these

34. A wire having a linear density of 0.1 kg/m is kept under a tension of 490 N. It is observed that it resonates at a frequency of 400 Hz. The next higher frequency is 450 Hz. Find the length of wire
    (1) 0.4 m (2) 0.7 m
    (3) 1.2 m (4) 1.3 m

35. Three coherent sound waves have amplitudes of 12 mm, 6 mm and 4 mm arrive at a given point with successive phase difference of $\frac{\pi}{2}$. The amplitude of resultant wave is
    (1) 7 mm (2) 10 mm
    (3) 4.8 mm (4) 14 mm

36. The speed of sound in hydrogen gas at 27°C is
    (1) 1520 m s^{-1} (2) 1670 m s^{-1}
    (3) 1320 m s^{-1} (4) 1830 m s^{-1}

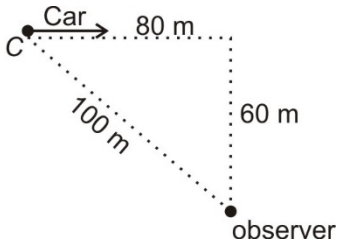
37. Third overtone of a closed organ pipe is in unison with fourth harmonic of an open organ pipe. The ratio of lengths of those pipes is
    (1) $\frac{5}{6}$ (2) $\frac{7}{8}$
    (3) $\frac{2}{3}$ (4) $\frac{4}{7}$

38. The string of a violin emits a note of frequency 600 Hz at its correct tension. Now the string is bit taut and produces 5 beats per second with a tuning fork of frequency 600 Hz. What is frequency of note emitted by this taut string?

(1) 605 Hz (2) 600 Hz

(3) 610 Hz (4) 595 Hz

39. A car is approaching a crossing at a speed of 20 m/s, sounds a horn of frequency 500 Hz when 80 m from crossing. Speed of sound in air is 330 m s^{-1}. What frequency is heard by an observer 60 m from the crossing on straight road which crosses the road at right angles as in the figure.



(1) 505 Hz (2) 510 Hz

(3) 525 Hz (4) 550 Hz

40. A window whose area is 2 m^2 opens on a street where the street noise results at the window an intensity level of 60 dB. What is the intensity of sound wave that enters through the window?

(1) 4 $\mu W\ m^{-2}$

(2) 1 $\mu W\ m^{-2}$

(3) 6 $\mu W\ m^{-2}$

(4) 12 $\mu W\ m^{-2}$

41. When sound wave travels from water to air, it

(1) Bends towards the normal

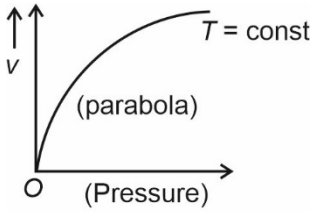
(2) Bends away from the normal

(3) May not emerge out

(4) May bend in any direction

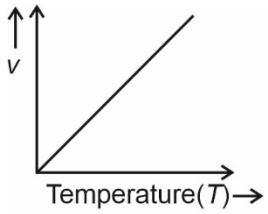
42. Which of the following graphs is correct?

(1)



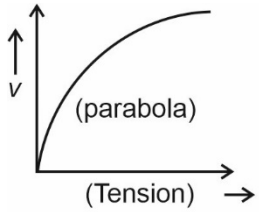
(v = velocity of sound)

(2)



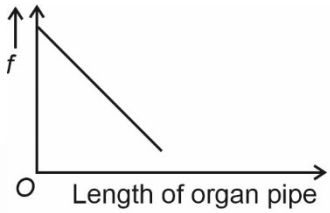
(Velocity of sound = v)

(3)



(v = Velocity of transverse wave in string)

(4)



(f = fundamental frequency of an organ pipe)

43. A sound wave travels with velocity of 300 m s^{-1}. through a gas. 9 beats are produced in 3 s, when two waves pass through it simultaneously. If one of wave has 2 m wavelength, the wavelength of other wave may be

(1) 2.04 m (2) 1.24 m

(3) 1.68 m (4) 1.32 m

44. A pulse of wave trains travels along a stretched string and reaches fixed end of the string. It will be reflected back with

(1) A phase change of 180° with velocity reversed

(2) Same phase as the incident pulse with no reversal of velocity

(3) A phase change of 180° with no reversal of velocity

(4) Same phase as the incident pulse but with reverse velocity

45. A string of mass 2.5 kg is under a tension of 200 N. The length of string is 20 m. If the transverse jerk is struck at one end of the string, the disturbance will reach other end in

(1) 1 s (2) 0.5 s

(3) 2 s (4) 2.5 s

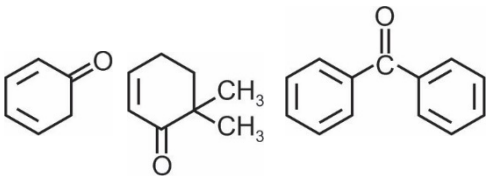
CHEMISTRY

46. The most stable carbocation among the following is

(1) $(CH_3)_3\overset{\oplus}{C}$ (2) $Ph-\overset{\oplus}{C}H_2$

(3) $CH_2=CH-\overset{\oplus}{C}H_2$ (4) $CH_2=\overset{\oplus}{C}H$

47. The compound(s) which show(s) keto-enol tautomerism is/are



I II III

(1) I only (2) II and III only

(3) I and II only (4) I, II and III

48. During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.2 g of the compound was neutralized by 5 mL of 1 M H_2SO_4. The percentage of nitrogen in the compound is

(1) 90% (2) 55%

(3) 70% (4) 62%

49. Select the incorrect statement.

(1) Electrophiles attack at high electron density sites

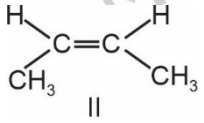
(2) Electrophiles are Lewis bases

(3) Generally electrophiles are electron deficient species

(4) SO_3 is an example of electrophile

50. Consider the given compounds

$(CH_3)_2C=C(CH_3)_2$ (I)



(II)

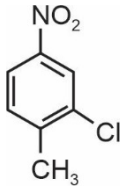
$(CH_3)_2C=CHCH_3$ (III)

The magnitude of enthalpy of hydrogenation of these compounds will be in the order

(1) II > I > III (2) I > III > II

(3) III > I > II (4) II > III > I

51. The IUPAC name of the given compound is



(1) 4-Methyl-5-chloro-nitrobenzene

(2) 2-Chloro-1-methyl-4-nitrobenzene

(3) 3-Chloro-4-methyl-1-nitrobenzene

(4) 6-Chloro-4-nitrotoluene

52. Which of the following compounds will exhibit cis-trans isomerism?

(1) 

(2) $CH_3CH=C(CH_3)_2$

(3) $PhCH_2-CH=CH_2$

(4) $Ph-CH=CH-Ph$

53. The general molecular formula, which represents the homologous series of alkanone is

(1) $C_nH_{2n+2}O$ (2) $C_nH_{2n}O$

(3) $C_nH_{2n-2}O$ (4) $C_nH_{2n+1}O$

54. $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OH$ and $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_3$ are

(1) Chain isomers (2) Position isomers

(3) Functional isomers (4) Metamers

55. In the given compound, the hybridization of C_1, C_3, C_4 and C_6 respectively are

$$\overset{(1)}{CH_3}-\overset{(2)}{C}\equiv\overset{(3)}{C}-\overset{(4)}{\overset{\overset{O}{\|}}{C}}-\overset{(5)}{CH_2}-\overset{(6)}{CN}$$

(1) sp^3, sp, sp^2, sp

(2) sp^3, sp, sp, sp^3

(3) sp^3, sp, sp^2, sp^2

(4) sp^2, sp, sp^2, sp^3

56. Species which has the highest –I effect among the following is

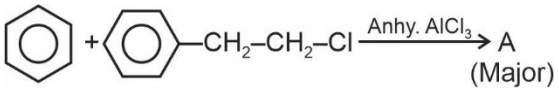
(1) – CN (2) – COOH

(3) $-NO_2$ (4) – OH

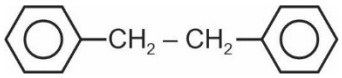
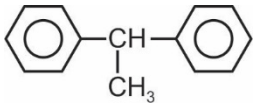
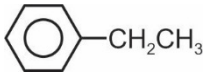
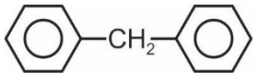
57. Which of the following species is aromatic in nature?

(1) 
(2) 
(3) 
(4) 

58. In the reaction,



A is

(1) 
(2) 
(3) 
(4) 

59. Maximum number of structural isomers possible for the molecular formula C_4H_8 is

(1) 4 (2) 5
(3) 6 (4) 3

60. The compound which is least reactive towards Friedel-Crafts reaction is

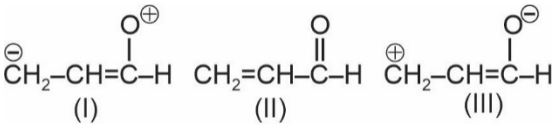
(1) Ethylbenzene (2) Chlorobenzene
(3) Nitrobenzene (4) Benzene

61. But-2-yne $\xrightarrow{X}$ cis-But-2-ene

The reagent (X) used in the above reaction is

(1) Na/C_2H_5OH
(2) Na/Liquid NH_3
(3) $LiAlH_4$
(4) $H_2/Pd/C$ (Lindlar's catalyst)

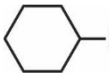
62. The correct order of stability of the following canonical forms is

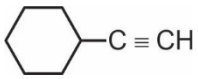
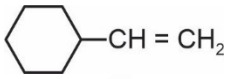
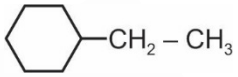


(1) I > II > III (2) II > III > I
(3) II > I > III (4) I > III > II

63. Heating sodium propanoate with sodalime gives

(1) Hexane
(2) Ethane
(3) 2, 3-dimethylbutane
(4) Butane

64. The product obtained on reaction of  with acetylene is

(1) 
(2) 
(3) 
(4) 

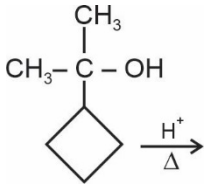
65. The compound which will not react with bromine water is

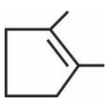
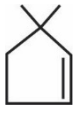
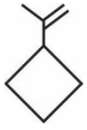
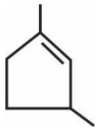
(1) 
(2) $PhCH_2CH = CH_2$
(3) $CH_3 - C \equiv CH$
(4) 

66. The most stable conformer of ethane –1,2-diol is

(1) Fully eclipsed form
(2) Gauche form
(3) Antiform
(4) Partially eclipsed form

67. Identify the major product formed in the following reaction



(1) 
(2) 
(3) 
(4) 

68. The products obtained on reductive ozonolysis of $CH_3 - \underset{\underset{CH_3}{|}}{C} = C = CH - CH_3$ are

(1) $CH_3CHO + CO_2 + CH_3CH_2OH$

(2) $CH_3 - CO - CH_2CH_3 + CH_3 - CH_2OH$

(3) $CH_3 - CO - CH_3 + CH_3CHO + CO_2$

(4) $CH_3 - CO - CH_3 + CH_3CHO + HCHO$

69. The technique used to separate chloroform and aniline is

(1) Distillation (2) Crystallisation

(3) Chromatography (4) Steam distillation

70. On reaction of sodium nitroprusside with sodium fusion extract, violet colour appears. It indicates the presence of which element in the sample?

(1) S (2) N

(3) Br (4) P

71. 'Blue baby syndrome' is caused by the excess of

(1) Lead in drinking water

(2) Nitrates in drinking water

(3) Sulphates in drinking water

(4) PAN in the atmosphere

72. n-Hexane on heating to 773 K at 10–20 atm pressure in presence of vanadium pentaoxide forms

(1) $CH_3 - \underset{\underset{CH_3}{|}}{CH} - \underset{\underset{CH_3}{|}}{CH} - CH_3$

(2) 

(3) $CH_2 = CH - CH_2 - CH_2 - CH_2 - CH_3$

(4) 

73. The alkene which gives only one product on reaction with HBr is

(1) 2, 3-dimethylbut-2-ene

(2) Pent-2-ene

(3) 2-methylbut-2-ene

(4) Hex-2-ene

74. Which of the following species adds in the first step during the addition of propene with ICl?

(1) Cl^+ (2) I^+

(3) Cl^- (4) I^-

75. Consider the following reaction,

$Ph - C \equiv C - CH_3 \xrightarrow{Hg^{2+}/H_3O^+} A\ (Major)$.

Identify A.

(1) $Ph - CH_2 - \underset{\underset{OH}{|}}{CH} - CH_3$

(2) $Ph - \underset{\underset{O}{||}}{C} - CH_2 - CH_3$

(3) $Ph - \underset{\underset{OH}{|}}{CH} - CH_2 - CH_3$

(4) $Ph - \underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}} - OH$

76. Lassaigne's test for nitrogen is negative for which of the following compounds?

(1) Hydrazine

(2) Amide

(3) Urea

(4) Ethylamine

77. During sulphonation of benzene, the electrophile is

(1) H_2SO_4

(2) SO_2

(3) HSO_4^-

(4) SO_3

78. Maximum prescribed concentration of Fe in drinking water is

(1) 0.2 ppm

(2) 0.02 ppm

(3) 5 ppm

(4) 0.5 ppm

79. Normally, the pH of rain water is

(1) 4.2

(2) 6.3

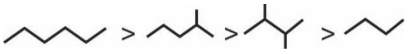
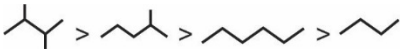
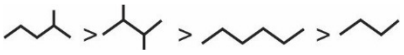
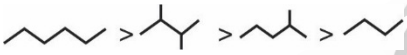
(3) 5.6

(4) 7.2

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80. Which of the following is not a green house gas?
(1) Nitrous oxide
(2) Methane
(3) Ozone
(4) Sulphur dioxide

81. Identify the incorrect statement.
(1) Ozone layer prevents the sun's harmful (UV) radiation from reaching the earth's surface
(2) The atmosphere that surrounds the earth has the same thickness at all heights
(3) Dinitrogen and dioxygen are the main constituents of air
(4) SO_2 is poisonous to both animals and plants

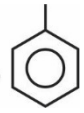
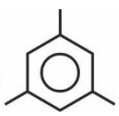
82. Correct order of boiling point for the given alkanes is
(1) 
(2) 
(3) 
(4) 

83. The major contributor(s) of acid rain is/are
(1) SO_2 (2) NO_2
(3) CO (4) Both (1) and (2)

84. Which of the following is incorrect regarding photochemical smog?
(1) It occurs in warm, dry and sunny climate
(2) Its main components results from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides
(3) It is called as reducing smog
(4) All of these

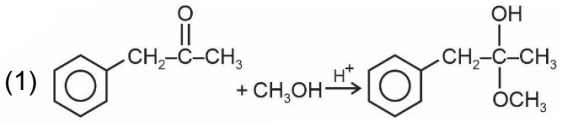
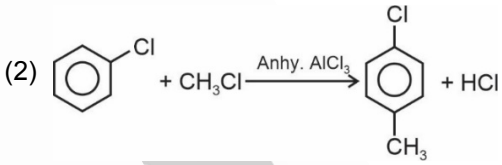
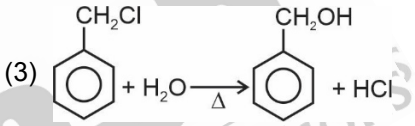
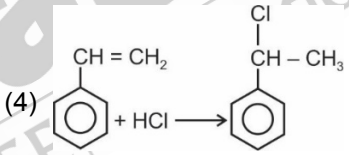
85. Consider the following reactions,

$$CaC_2 + 2H_2O \longrightarrow A \xrightarrow[873\,K]{\text{Red hot iron tube}} B\,.$$

A and B respectively are
(1) C_2H_4, 
(2) C_2H_2, 
(3) C_2H_2, 
(4) C_2H_6, 

86. The number of σ and π bonds in phenanthrene molecule respectively are
(1) 24, 5
(2) 28, 7
(3) 19, 5
(4) 26, 7

87. For which of the following compounds, Kjeldahl's method is not applicable?
(1) Azobenzene
(2) Nitrobenzene
(3) Pyridine
(4) All of these

88. Which one of the following is a nucleophilic substitution reaction?
(1) 
(2) 
(3) 
(4) 

89. The alkane which on monochlorination gives maximum number of products (structural isomers) is
(1) 2-2-dimethylpropane
(2) 2-methylpropane
(3) 2-methylbutane
(4) Pentane

90. Which of the following is a pesticide?
(1) Sodium chlorate
(2) Dieldrin
(3) Sodium Arsinite
(4) Potassium permanganate

BOTANY

91. Which of the following statements is **not** correct with reference to respiration in plants?
(1) Plants like animals have special systems for gaseous exchange
(2) Stomata, barks and lenticels are meant for gas exchange
(3) Gaseous exchange in plants is by diffusion
(4) All living cells in plant have their surfaces exposed to air

92. Which of the following is the favoured substrate for respiration?
(1) Organic acids
(2) Glucose
(3) Fats
(4) Proteins

93. How many molecules of CO_2 are evolved when two molecules of tripalmitin are used as respiratory substrate to measure the RQ value?
(1) 145 molecules (2) 45 molecules
(3) 102 molecules (4) 6 molecules

94. Identify the cycle operating in the matrix of the mitochondria.
(1) Glycolysis
(2) Tricarboxylic acid cycle
(3) Calvin cycle
(4) Hatch-Slack cycle

95. Which of the following is pair of interconvertible triose phosphates?
(1) DHAP – PGAL
(2) PGAL – PGA
(3) PGAL – Pyruvic acid
(4) Pyruvic acid – Acetyl CoA

96. Which process provides raw material for oxidation of food?
(1) Respiration
(2) Photorespiration
(3) Photosynthesis
(4) Transpiration

97. In which of the following conversions, there is change in carbon number from reactant to product?
(1) Conversion of PEP to pyruvic acid
(2) Conversion of citric acid to oxalosuccinic acid
(3) Conversion of acetaldehyde to ethyl alcohol
(4) Conversion of α-ketoglutaric acid to succinic acid

98. The aerobic respiration differs from fermentation, as the former process
(1) Accounts only for a partial breakdown of glucose
(2) Has a net gain of 2 ATP from a glucose molecule
(3) Oxidised NADH to NAD^+ vigorously
(4) Involves conversion of 3-phosphoglyceric acid to 2-phosphoglyceric acid

99. When glucose is used as respiratory substrate the efficiency of aerobic respiration comes out as
(1) 95%
(2) 75%
(3) 45%
(4) 25%

100. EMP pathway and C_3 pathway occurs respectively in
(1) Mitochondrial matrix and cytoplasm
(2) Cytoplasm and chloroplast stroma
(3) Cytoplasm and mitochondrial matrix
(4) Chloroplast stroma and cytoplasm

101. The reactions catalysed by pyruvic dehydrogenase require the participation of several co-enzymes like
(1) ATP and GTP
(2) NAD^+ and Co-enzyme A
(3) Mg^{+2} and $FADH_2$
(4) Co-enzyme A and $NADP^+$

102. What is the net ATP gained when 4 molecules of lactic acid are formed from glucose molecules?
(1) 38 (2) 36
(3) 4 (4) 8

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103. There are 'A' major way(s) in which different cells handle pyruvic acid produced by glycolysis. 'A' represents.
(1) Three (2) Two
(3) One (4) Six

104. Which of the following statements is **wrong** for both lactic acid and alcoholic fermentation?
(1) Not much energy is released
(2) Processes are hazardous
(3) NADH + H^+ is oxidised to NAD^+
(4) More than 50% percent of energy in glucose is released

105. The deficiency symptoms of N, Mg and K are first visible in the senescent leaves because
(1) They are immobile elements
(2) They are mobile elements
(3) They are critical elements
(4) They are micronutrients

106. CO_2 is **not** released during which of the given processes?
(1) Alcoholic fermentation
(2) Link reaction
(3) Glycolysis
(4) Krebs cycle

107. Read the statements A and B and choose the **correct** option.
A : Uptake of various mineral ions into the symplast of the cells involves active transport.
B : Some minerals are absorbed passively also.
(1) A and B both are correct
(2) A and B both are incorrect
(3) Only A is correct
(4) Only B is correct

108. Which essential element is absorbed in anionic form?
(1) Boron (2) Copper
(3) Zinc (4) Molybdenum

109. The manganese activates the enzymes involved in all of the given processes, **except**
(1) Photosynthesis (2) Respiration
(3) Nitrogen metabolism (4) Pollen germination

110. Which of the following characteristics is **correct** for calcium?
(1) It is relatively immobile
(2) It is transported out of the mature organs
(3) Deficiency symptoms of calcium tend to appear first in senescent organs
(4) It is a trace element

111. Which of the given is a free living anaerobic nitrogen fixer?
(1) *Beijerinckia* (2) *Rhodospirillum*
(3) *Azotobacter* (4) *Frankia*

112. Both *Rhizobium* and *Frankia* are
(1) Symbionts in leguminous plants
(2) Parasites on roots of leguminous plants
(3) Free living but can fix atmospheric nitrogen as symbionts
(4) Chemoautotrophs

113. Which of the following statements is **correct**?
(1) Purple and green sulphur bacteria show oxygenic photosynthesis
(2) Sucrose is a six carbon sugar
(3) In bacterial photosynthesis, the electron donor is H_2S
(4) The O_2 evolved during photosynthesis process comes from H_2S in green plants

114. "Glucose is usually stored as starch in plants." It was first established by
(1) Van Niel
(2) Julius Von Sachs
(3) Engelmann
(4) Priestley

115. The stromal lamella of chloroplast connects
(1) Two adjoining grana
(2) Two adjoining starch granules
(3) Granum with inner membrane
(4) Lipid droplet with starch granule

116. PS II and PS I have absorption maxima at
(1) 600 nm and 700 nm respectively
(2) 700 nm and 680 nm respectively
(3) 700 nm and 600 nm respectively
(4) 680 nm and 700 nm respectively

117. 8 molecules of water split into
(1) $2H^+$, O_2 and $4e^-$
(2) $4H^+$, O_2 and $8e^-$
(3) $8H^+$, $2O_2$ and $8e^-$
(4) $16H^+$, $4O_2$ and $16e^-$

118. By using radioactive ^{14}C, Melvin Calvin discovered that the first CO_2 fixation product was a
(1) 5-carbon compound
(2) 6-carbon compound
(3) 3-carbon organic acid
(4) 5-carbon keto sugar

119. In Hatch and Slack pathway, the primary CO_2 acceptor is
(1) 5-carbon keto sugar
(2) 5-carbon organic acid
(3) 4-carbon organic acid
(4) 3-carbon organic acid

120. Chemiosmosis requires
(1) Only outer membrane of chloroplast and mitochondria
(2) Transmembrane channels of the CF_1 of the ATP synthase
(3) A proton gradient across the membrane
(4) Positive hydrostatic pressure in xylem

121. Which of the following stages of the Calvin cycle involves utilisation of ATP and NADPH molecules both?
(1) Carboxylation (2) Reduction
(3) Decarboxylation (4) Regeneration

122. The leaves of C_4-plant differ from the C_3-plant in having
(1) Small cells around the vascular bundle called bundle sheath cells
(2) Single layer of bundle sheath cells with large intercellular spaces present around the vascular bundle
(3) Large intercellular space present between the bundle sheath cells
(4) Large number of agranal chloroplasts in wreath cells

123. The first C_4-acid formed in the mesophyll cells of C_4-plant is
(1) Oxaloacetic acid
(2) Malic acid
(3) α-Ketoglutaric acid
(4) Aspartic acid

124. Regeneration of RuBP is crucial for continuity of CO_2 fixation. Regeneration of 6 molecules of RuBP requires
(1) 18 ATP (2) 12 ATP
(3) 6 ATP (4) 24 ATP

125. How many ATP and NADPH are required in *Sorghum*, for fixation of 18 molecules of CO_2?
(1) 90 ATP and 12 NADPH
(2) 36 NADPH and 90 ATP
(3) 90 NADPH and 36 ATP
(4) 30 ATP and 12 NADPH

126. At higher CO_2 concentration, the increased rates of photosynthesis leading to higher productivity has been used for greenhouse crops such as tomatoes and bell pepper. The above result reflects that the tomato and bell pepper are
(1) CAM plants (2) C_3-plant
(3) C_4-plant (4) C_2-plant

127. Which of the following is **wrong** for reductive amination?
(1) It occurs in the presence of glutamate dehydrogenase
(2) Ammonium ion directly combines with α- ketoglutaric acid
(3) A oxidised coenzyme is required
(4) Water molecule is produced

128. Which one of the following elements is **not** in the list of essential elements, but beneficial in some plants?
(1) Zinc (2) Molybdenum
(3) Boron (4) Sodium

129. Moll's half leaf experiment showed that the
(1) CO_2 is essential for photosynthesis
(2) Light is essential for photosynthesis
(3) Starch is first visible product of photosynthesis
(4) O_2 is evolved during light reaction

130. How many complexes are involved in mitochondrial ETS?
(1) Zero (2) Two
(3) Three (4) Four

131. Conversion of glucose to glucose-6-phosphate is catalysed by
(1) Enolase (2) Hexokinase
(3) Pyruvate kinase (4) PFK

132. Choose the **incorrect** match
(1) Leghaemoglobin – Oxygen scavenger
(2) Nitrogenase – Manganese and iron
(3) *Nostoc* – Heterocyst
(4) Auxin – Zinc

133. Denitrification is carried by which of the given pair of bacteria?
(1) *Pseudomonas* and *Thiobacillus*
(2) *Nitrosomonas* and *Anabaena*
(3) *Bacillus* and *Spirillum*
(4) *Pseudomonas* and *Nitrobacter*

134. More than 13 percent of alcohol in a beverage is achieved through
(1) Purification of alcohol
(2) Fermentation by *Saccharomyces*
(3) Distillation of ethyl alcohol
(4) Bacterial fermentation

135. The following are the significance of hydroponics, **except**
(1) Commercial production of lettuce and tomato
(2) It controls soil-borne pathogens
(3) Identifying the deficiency symptom of an essential element
(4) It is very cost effective technique for high yield

ZOOLOGY

136. Identify the statements as true **(T)** or false **(F)** and choose the **correct** option.
(i) Cilia help in locomotion in *Paramecium*
(ii) *Hydra* uses tentacles for capturing of prey and also for locomotion
(iii) Female reproductive tract in humans exhibit flagellar movement to propel the ovum
(iv) Macrophages and leucocytes of humans are specialised cells that exhibit amoeboid movement by forming pseudopodia

	i	**ii**	**iii**	**iv**
(1)	T	T	F	F
(2)	T	F	T	F
(3)	T	T	F	T
(4)	T	T	T	F

137. Which of the following statements is **incorrect** w.r.t. cardiac muscle fibres?
(1) Exclusively present in myocardium of heart
(2) Striated in appearance
(3) They are involuntary in nature as the nervous system does not control their activities directly
(4) Their cells never assemble in a branching pattern

138. Compare to visceral muscle fibres the skeletal muscle fibres
(1) Assist in the transport of food through the digestive tract by peristalsis
(2) Involved in locomotory actions and changes of body postures
(3) Striated in appearance and involuntary in nature
(4) Are located in the inner walls of hollow visceral organs of the body

139. Read the following statements carefully w.r.t. skeletal muscle and choose the **correct** answer.
A : Each fascicle consist of numerous muscle fibres
B : All fascicles of a muscle are held together by fascia
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

140. Select the **incorrect** statement w.r.t. A band
(1) It appears dark
(2) Its middle part contains H zone which consist of only thin filaments
(3) It consists of both actin and myosin filaments and actin filament overlap the myosin filaments on either edges
(4) Its length remain same during muscle contraction

141. For muscle contraction, ATPase activity is present in
(1) LMM (2) HMM
(3) Tail of meromyosin (4) G-actin

142. Choose the option with all **incorrect** events occurring during muscle contraction.
A. Size of sarcomere – Decreases gradually and ultimately become less than size of the dark band
B. Size of I band – Increases gradually
C. Size of H zone – Remains same as that of dark band
(1) Only A and B (2) Only B and C
(3) Only C (4) A, B and C

143. Red muscle fibres
(1) Contain less quantity of myoglobin
(2) Possess high amount of sarcoplasmic reticulum
(3) Only depend on anaerobic process for energy
(4) Can utilise the large amount of oxygen for ATP production due to plenty of mitochondria

144. Zygomatic bones are included in all **except**
(1) Skull
(2) Axial skeleton
(3) Appendicular skeleton
(4) Facial bones

145. In human skeleton, the number of limb bones are
(1) 20 times to the number of girdle bones
(2) 60 times to number of ear ossicles
(3) 30 times to number of cranial bones
(4) 2 times to number of bones of axial skeleton

146. A paired structure present in a bone of human skull forms condyloid joint with bone of vertebral column. The structure is
(1) Hyoid bone
(2) Occipital condyle
(3) Odontoid peg
(4) Vertebra prominens

147. Which of the following is a large triangular flat bone of appendicular skeleton present between the second and the seventh ribs?
(1) Humerus (2) Scapula
(3) Clavicle (4) Pubis

148. Choose the **correct** statement.
(1) Thoracic vertebrae and ribs together with sternum form rib cage
(2) Last 2 pairs of ribs (11th and 12th) called vertebrochondral ribs
(3) All ribs articulate with sternum either directly or indirectly
(4) Sternum articulates with all bones of pectoral girdle

149. How many structures from given below in the box are included in girdles of appendicular skeleton?

Coxal bone, Ileum, Sacrum, Femur, Clavicle, Patella, Tibia

(1) Two
(2) Three
(3) Four
(4) Five

150. Which of the following joint is present between 1st and 2nd vertebra of vertebral column?
(1) Ball and socket joint
(2) Pivot joint
(3) Gliding joint
(4) Saddle joint

151. The joints which allow considerable movements and play a significant role in locomotion are
(1) Ball and socket joint, joint between skull bones
(2) Hinge joint, ball and socket joint
(3) Joint between adjacent vertebrae, saddle joint
(4) Gliding joint, joint between skull bones

152. Choose the **mismatch** w.r.t. disorders of muscular/skeletal system.
(1) Muscular dystrophy – Progressive degeneration of only visceral muscle
(2) Tetany – Occurs due to low Ca^{+2} in body fluids
(3) Gout – Occurs due to disturbance of metabolism of purine
(4) Osteoporosis – Decreased levels of estrogen is a common cause in females

153. Select the **correct** set of bones which are present only in lower limbs.
(1) Tibia, radius (2) Humerus, tibia
(3) Coxal bone, femur (4) Fibula, tarsals

154. All true ribs are A attached to the thoracic vertebrae and B connected to the sternum with the help of hyaline cartilage.
Choose the option that **correctly** fill the blanks A and B respectively.
(1) Ventrally, Medially
(2) Dorsally, Ventrally
(3) Laterally, Ventrally
(4) Laterally, Dorsally

155. The term bicephalic is applicable for
(1) Sternum (2) Ribs
(3) Thoracic vertebrae (4) Sacrum

156. Match the column I with column II and select the **correct** option.

Column I	Column II
(i) Ankle bones	(a) Help in articulation of limbs with axial skeleton
(ii) Neural canal	(b) 14 in number
(iii) Girdle bones	(c) Central hollow portion in each vertebra
(iv) Acromian process	(d) Present in scapula

(1) (i) – b, (ii) – c, (iii) – a, (iv) – d
(2) (i) – b, (ii) – c, (iii) – d, (iv) – a
(3) (i) – b, (ii) – a, (iii) – c, (iv) – d
(4) (i) – a, (ii) – b, (iii) – c, (iv) – d

157. Which of the following primarily constitutes the main framework of the trunk?
(1) Skull (2) Vertebral column
(3) Sternum (4) Ribs

158. Read the following statements.
(i) Each middle ear contains ear ossicles
(ii) Parietal and ethmoid both are paired bones of cranium
(iii) Our skeletal system consist of a framework of bones and a few cartilages.
Choose the option with only **correct** statements.
(1) (i) and (ii) (2) (i) and (iii)
(3) (ii) and (iii) (4) (i), (ii) and (iii)

159. Ischium bone is a structural constituent of
(1) Pectoral girdle (2) Coxal bone
(3) Axial skeleton (4) Lower limbs

160. Select the bone of appendicular skeleton which also articulates directly with axial skeleton
(1) Pubis (2) Sternum
(3) Scapula (4) Clavicle

161. Which of the following is **incorrect** w.r.t. resting membrane potential in a nerve cell?
(1) More permeability of Na^+ compare to K^+ of axonal membrane
(2) Active transport of $3Na^+$ outwards for $2K^+$ into the nerve cell by sodium – potassium pump
(3) Polarised condition across the axonal membrane
(4) All voltage gated channels present on axonal membrane remain close

162. Cerebral aqueduct is a structural constituent of
(1) Forebrain
(2) Hindbrain
(3) Midbrain
(4) Medulla oblongata

163. Choose the **odd** one w.r.t. limbic system.
(1) Amygdala (2) Hippocampus
(3) Hypothalamus (4) Cerebellum

164. Site for processing of hearing and speech is present in
(1) Forebrain
(2) Midbrain
(3) Hindbrain
(4) PNS

165. Select the **incorrect** statement for efferent neuron which participates in knee jerk reflex action.
(1) Emerges from white matter of spinal cord
(2) Makes synapse directly with afferent neuron
(3) Ends at motor end plate of effector
(4) Included in PNS

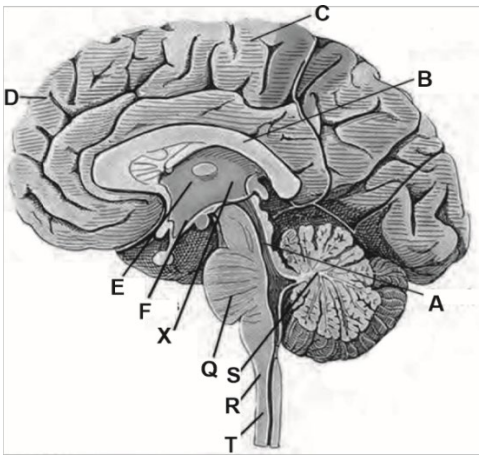
166. In cerebral hemispheres,
(1) Grey matter surrounds the white matter
(2) White matter surrounds the grey matter
(3) Prominent folds are present in white matter
(4) Two ventricles are found in each hemisphere

167. In each middle ear of mammals, the ear ossicle stapes is found attached with
(1) Round window of the vestibule
(2) Oval window of the cochlea
(3) Tympanic membrane
(4) Temporal bone of skull

168. Cochlea of human's ear contains Reissner's membrane, forms the boundary between
(1) Scala vestibuli and scala media
(2) Scala media and scala tympani
(3) Scala vestibuli and macula
(4) Crista ampullaris and macula

169. Which among these is **not** a sensory structure?
(1) Crista
(2) Arachnoid
(3) Macula
(4) Retina

170. Identify the figure below given and select the labelled structures present in brain stem.



(1) X, S, T (2) P, Q, R
(3) X, Q, R (4) E, Q, F

171. Choose the **incorrect** statement w.r.t. human brain.
(1) Centres for respiration control, cardiovascular reflex and gastric secretions are present in hindbrain
(2) Corpus callosum is a band of white nerve fibres exclusively present in forebrain between cerebral hemispheres
(3) Association areas of cerebral cortex is responsible for complex functions like memory and communications
(4) Hypothalamus controls body temperature, urge of eating and also it is a major coordination centre for sensory and motor signaling

172. Which of the following is exclusive for PNS?
(1) Nodes of Ranvier
(2) Schwann cells
(3) Myelinated nerve fibres
(4) Multipolar neuron

173. In visceral neural system, impulses travel
(1) Only from CNS to viscera
(2) Only from viscera to CNS
(3) From CNS to viscera and vice versa
(4) From CNS to skeletal muscles

174. Parasympathetic nervous system influences all **except**
(1) Secretion of digestive juices
(2) Secretion of adrenal cortex
(3) Secretion of saliva
(4) Rate of heart beat

175. Which of the following is **correct** w.r.t. differences between fovea centralis and blind spot of retina of human eye?

		Fovea centralis	**Blind spot**
(1)	Location	Lateral to the blind spot, at the posterior pole of the eye	Medial to and slightly above the posterior pole of the eye ball
(2)	Visual activity	Low	High
(3)	Photoreceptor cells contained	Both rods and cones	Only rods
(4)	Function	Vision in poor light	Vision in bright light

176. Crista ampullaris that acts as sensory receptor for sensing the change in the position of head arises from

(1) Ampulla of semi-circular canals

(2) Macula

(3) Saccule and utricle

(4) Organ of Corti

177. Scotopic vision is the function of A which are not present at B in retina, where optic nerves leave the eye ball.

Here A and B are

	A	**B**
(1)	Rods	Fovea centralis
(2)	Rods	Blind spots
(3)	Cone	Fovea centralis
(4)	Cone	Blind spots

178. Vitreous chamber is filled with a

(1) Aqueous humor

(2) Thin watery fluid

(3) Transparent gel

(4) Saline solution

179. Read the following statements w.r.t. pupil and select the **incorrect** statement.

(1) It is an aperture present in front of the lens

(2) It is surrounded by the Iris

(3) Its diameter is regulated by the muscle fibres of ciliary body

(4) Light rays enter into eye though it

180. Select the structure present in bipolar neuron but **not** in unipolar neuron

(1) Cell body (2) Axon

(3) Dendrites (4) Soma

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